AI-1 Spring 26 Project Proposal

Team Members

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Idea Short Name (sudoku solver)

Idea Description

Intelligent Sudoku Solver: An Application of Constraint Satisfaction and Depth-First Search

Agent Architecture

☐ Simple Reflex Agent

☐ Model-Based Reflex Agent

√ Goal-Based Agent

☐ Utility-Based Agent

Problem-Solving Framework

☐ Classical Search

☐ Local Search (Optimization)

☐ Adversarial Search (Two-Player Games)

√ Constraint Satisfaction Problem

Proposed Algorithm(s)

CSP (Constraints Satisfaction Problem) **DFS (Depth First Search)**

PEAS

Performance Measures	Solve the puzzle correctly within the constraints, minimize solving time (number of backtracks)
Environment	A 9x9 grid with some pre-filled cells
Actuators	Filling in a digit (1-9) into an empty cell
Sensors	Reading the current grid (current cell values and empty cells)

Environment Properties

√ Fully Observable ☐ Partially Observable	√ Deterministic ☐ Stochastic ☐ Strategic	√ Discrete ☐ Continuous
√ Single-Agent ☐ Multi-Agent	√ Sequential ☐ Episodic	√ Static ☐ Dynamic